Stable vertices and the core of a matching do not depend on the reduction process

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doi:10.5802/alco.351; CC BY 4.0

Editorial note. The counted statement is the author's labelled proposition environment at publisher-source.tex lines 846-849; its proof environment opens at line 850, immediately after \end{proposition}, so the pairing is by adjacency and the retained proof range 850-853 is the whole environment.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label prop:core well defined.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial note. The retained preamble is the editorially prepared standalone preamble of this source, which already contains the document class, the ams packages, the native theorem-environment family and the native author macros.

Editorial scope. One original result of the pinned publisher article, counted once: well-definedness of the stable vertices and of the core of a matching (author label prop:core well defined).

Allowed standard prerequisites (editorial).

- Short chord and reduction process. A chord is short when it has no vertex strictly between its endpoints; a reduction process removes short chords one after another and pushes the remaining vertices together.. It is the process whose results the proposition shows to be independent of the choices made.
- Stable vertex and core. A vertex is stable when it is not removed by any reduction process, and the core of a matching is the matching induced on the stable vertices.. They are the two objects the proposition proves well defined.

Proposition 0.1. Given a matching, its stable vertices and its core are well-defined.

Proof. Let $m \in \mathcal{M}_N$ be a matching. Denote the chords that are removed during a given reduction process of m by $e_1, \dots, e_k$, and denote $e_\ell = (i_\ell, j_\ell)$. Moreover, denote the chords that are removed during another reduction process by $e'_1, \dots, e'_{k'}$, and denote $e'_\ell = (i'_\ell, j'_\ell)$. It suffices to prove that $e_\ell \in \{e'_1, \dots, e'_{k'}\}$ for all $1 \le \ell \le k$ (i.e. every chord that is removed during the first reduction process is removed during the second process as well). Assume by contradiction that $e_\ell \notin \{e'_1, \dots, e'_{k'}\}$ for some ℓ, and denote by ℓ_0 the minimal such ℓ. That is, $e_1, \dots, e_{\ell_0 - 1} \in \{e'_1, \dots, e'_{k'}\}$. Removing the chords $e_1, \dots, e_k$ from m constitutes a valid reduction process, so the chord e_{ℓ_0} becomes short before it is removed. That is, $i \in \{i_1, j_1, \dots, i_{\ell_0 - 1}, j_{\ell_0 - 1}\}$ for all $i_{\ell_0} < i < j_{\ell_0}$. Therefore, the chord e_{ℓ_0} is a short chord of the core obtained by removing $e'_1, \dots, e'_{k'}$ of m, contradicting

the requirement that the reduction process continues until there are no short chords remaining. Therefore, we may conclude that if a chord is removed during a reduction process then it is removed during any reduction process, and $\mathrm{core}(m)$ and $\mathrm{Stable}(m)$ are well-defined. □